International Vaccine Access Center
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Joshua C. Macdonald

Curriculum Vitae

Research Profile

Computational scientist developing modeling, inference, and scientific software for global health research. My work spans human, animal, and environmental health systems, integrating mechanistic modeling, Bayesian and statistical inference, uncertainty quantification, and scientific computing to study partially observed processes under sparse, heterogeneous data. Current application areas include infectious disease dynamics, host-pathogen and comparative health systems, environmental hazards, and computational tools for forecasting and decision support.

Education

2022 **PhD, Mathematics**, *University of Louisiana at Lafayette*, Lafayette, Louisiana, USA, Advisor: Dr. Hayriye Gulbudak
Dissertation: *Population Dynamics Under Disturbance in Ecology and Epidemiology*

2020 **MS, Mathematics**, *University of Louisiana at Lafayette*, Lafayette, Louisiana, USA

2016 **BA, Mathematics**, *University of North Carolina Asheville*, Asheville, North Carolina, USA

2013 **BA, Archaeology**, *University of North Carolina Greensboro*, Greensboro, North Carolina, USA

Appointments

2024–Present **Postdoctoral Scholar**, *International Vaccine Access Center, Department of International Health (Global Disease Epidemiology and Control), Johns Hopkins Bloomberg School of Public Health*, Baltimore, Maryland, USA
Mentor: Dr. Shaun Truelove. Research focus: infectious disease modeling and inference, scientific computing, decision support systems, and research software engineering.

2022–2024 **Zuckerman STEM Leadership Postdoctoral Fellow**, *School of Zoology, Tel Aviv University*, Tel Aviv, Israel
Mentor: Dr. Yoav Ram. Research drew directly on archaeological training in the study of material culture diversity and cultural transmission, applying Bayesian dimensionality reduction and multivariate data analysis to cultural datasets.

Peer-Reviewed Publications

Published

2024 **Macdonald, J.C., Gulbudak, H., Beechler, B., Gorsich, E., Gubbins, S., Perez, E., & Jolles, A.**, *Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo*, *The American Naturalist*, doi:10.1086/730703

2024 **Macdonald, J.C., Blanco-Portillo, J., Feldman, M.W., & Ram, Y.**, *Cultural transmission, networks, and clusters among Austronesian-speaking peoples*, *Evolutionary Human Sciences*, 6, e45, doi:10.1017/ehs.2024.45

2023 **Macdonald, J.C. & Gulbudak, H.**, *Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms*, *Journal of Mathematical Biology*, 87, 45, doi:10.1007/s00285-023-01969-7

2022 **Browne, C.J., Gulbudak, H., & Macdonald, J.C.**, *First and second wave outbreak size with reactive quarantine measures: a COVID-19 model applied to an epidemic in China*, *Journal of Theoretical Biology*, 538, 110919, doi:10.1016/j.jtbi.2021.110919

2021 **Macdonald, J.C., Browne, C.J., & Gulbudak, H.**, *Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates*, *Royal Society Open Science*, 8, 210227, doi:10.1098/rsos.210227

Under Review

2026 **Atkins, S., Gulbudak, H., Welker, S., Smith, H., Macdonald, J.C., Beechler, B.R., & Jolles, A.E.**, *An immuno-epidemiological model of foot-and-mouth disease in African buffalo with carrier transmission*, Under review
Submitted to *Bulletin of Mathematical Biology*.

Preprints and Manuscripts in Revision

2024 **Macdonald, J.C., Blanco-Portillo, J., Feldman, M.W., & Ram, Y.**, *Bayesian estimation of the number of significant principal components for cultural data*, Preprint, doi:10.48550/arXiv.2409.12129

Manuscripts in Preparation

2026 **Macdonald, J.C.**, *Lawful learning: a scientific machine learning approach to dynamical systems inference*, In preparation for *SIAM Journal on Scientific Computing*

2026 **Macdonald, J.C.**, *Structured observables: a physics-based approach to dynamical systems inference*, In preparation for *SIAM Journal on Applied Dynamical Systems*

2026 **Roosa, K., White, L.A., Leon, T.M., Macdonald, J.C., Srivastava, A., & Truelove, S.**, *Shapelet analysis of infectious disease forecasts reveals heterogeneity across epidemic phases*, In preparation for *Epidemics*

Research Support

CDC-CFA (60% FTE)

2023–Present **Cooperative Agreement CDC-RFA-FT-23-0069 (NU38FT000012)**, *Postdoctoral Scholar*, Directors: Justin Lessler, Kimberly Powers, and Shaun Truelove, Title: Atlantic Coast Center for Infectious Disease Dynamics and Analytics (ACCIDDA)
Insight Net center for infectious disease forecasting and outbreak analytics. Research role: operational modeling, forecasting pipeline development, and computational infrastructure.

USDA (40% FTE)

2025–2028 **USDA-NIFA AFRI US–UK Collaboration (Award #1032897)**, *Collaborator*, US PI: Anna Jolles; linked UKRI award BB/Z516934/1 with Emma Thomson, Teresa Lambe, Roman Biek, and Georgina Limon-Vega, Title: Quiescence to emergence: Does nairovirus diversity loss increase disease transmission, severity and the risk of emergence?
Research role: mathematical model parameterization, data analysis, and advising on mathematical and statistical methods.

2025–Present **USDA-ARS Cooperative Agreement (3022-32000-021-018-S)**, *Collaborator*, Title: Investigating the Epidemiology of CCHFV in an Endemic Setting: Uganda
Research role: mathematical model parameterization, data analysis, and advising on mathematical and statistical methods.

Other

2022–2027 **NSF EEID / UKRI UK–US Collaboration Grant ($4.05M / 5 years)**, *Senior Personnel (from 2024)*, PIs: Brianna Beechler, Roman Biek, Hayriye Gulbudak, Erin Gorsich, Simon Gubbins, Anna Jolles, and Jan Medlock, Title: Multi-scale infection dynamics from cells to landscapes: FMD in African buffalo

Research role: mathematical model development and parameterization of within-host viral and immune dynamics.

2022–2024 **Zuckerman Postdoctoral Scholarship ($120K total)**, *Principal Investigator*, Title: Co-evolution of Cultural and Genetic Traits

Competitive postdoctoral fellowship; authored and defended research proposal.

Pending

In review **National One Health Mission India ($1M / 3 years)**, *Co-Investigator*, Title: Development of an Integrated One Health Disease Modelling and Digital Health Program in Human, Livestock, and Wildlife

Research role: mathematical model parameterization and data analysis; mechanistic modeling for HPAI and anthrax in India; stakeholder training in modeling; and supervision of postdoctoral and doctoral trainees.

Operational Modeling

Scenario Modeling Hubs

Multi-model scenario projections produced by the Scenario Modeling Hubs are used in part by the CDC Center for Forecasting and Outbreak Analytics (CDC-CFA) and the Advisory Committee on Immunization Practices (ACIP) to inform vaccine policy and pandemic preparedness decisions.

2024–Present **Flu Scenario Modeling Hub**, *ACCIDDA*, Lead Model Developer

Operational modeling and forecasting pipeline development. Leading scientific overhaul of ACCIDDA flu model in advance of 2026/27 operational round.
github.com/midas-network/flu-scenario-modeling-hub

2025 **COVID-19 Scenario Modeling Hub**, *ACCIDDA*, Support

Data cleaning and assimilation in support of ACCIDDA's health heterogeneities COVID-19 round.
github.com/midas-network/covid19-scenario-modeling-hub

Other Operational Modeling

2026–Present **Hib Vaccination Modeling (Navajo Nation)**, *ACCIDDA*

Technical supervisor managing PhD students in implementation of an age- and immune-status-structured Hib model for the Navajo Nation to evaluate the impact of long-running vaccination programs.

Research Software

Lead Developer

2026–Present **OP System (Operator-Partitioned System)**, *ACCIDDA*

Declarative governing equation specification compiler for structured dynamical systems (e.g., age-structured epidemic models, size-structured plankton models). Transforms YAML model definitions into callable numerics consumed by OP Engine or other solvers; designed to eliminate bookkeeping errors and reduce the barrier to simulation of complex models.
github.com/ACCIDDA/op_system

2026–Present **pp-eigentest**

Posterior predictive eigenvalue testing framework for determining signal rank in covariance and Gram matrices. Supports ordered hypothesis testing, family-wise error-rate control, and false discovery rate control. In development; pre-release.

2025–Present **OP Engine (Operator-Partitioned Engine)**, *ACCIDDA*

Numerical simulation engine for mechanistic models of biological and epidemiological systems. Supports ODE, PDE, and hybrid operator-splitting solvers with pluggable backends.
github.com/ACCIDDA/op_engine

2024–Present **vbpca-py**
Variational Bayesian PCA for incomplete data with native missing-data handling, uncertainty quantification, and C++-accelerated kernels; compatible with `scikit-learn`.
pypi.org/project/vbpca-py/ doi:10.5281/zenodo.19389250

Collaborative Infrastructure

2025–Present **FlepiMoP2 (Flexible Epidemic Modeling Pipeline 2)**, *ACCIDDA*
Core contributor to the redesign of a modular epidemic modeling and simulation pipeline for forecasting and scenario analysis.
Documentation

2025–Present **TabularEpi**, *UNC / ACCIDDA*
Rule-based tabular epidemic simulator: ingests DataFrames, performs internal numerics in NumPy/SciPy, and outputs state as DataFrames. Supports mixed deterministic-stochastic models with custom transition rules. Conducted systematic performance benchmarking that identified critical architectural bottlenecks, motivating an ongoing refactor to a vectorized backend.

2024–2025 **FlepiMoP (Flexible Epidemic Modeling Pipeline)**, *Hopkins IDD / ACCIDDA*
Developed modular ODE solver interfaces enabling integration of external simulation and inference models; implemented a PyMC-based Bayesian inference module for operational influenza modeling; and conducted benchmarking that identified vectorization-driven performance gains of 5–20×, informing the architecture of FlepiMoP2.

Presentations

Invited Talks and Seminars

2026 **Cross-Scale Feedback Motifs: Structure-Preserving Models and Computational Tools for Complex Systems**
Biological Oceanography Seminar, Woods Hole Oceanographic Institution, Woods Hole, Massachusetts

2026 **Decision-Support Modeling for One Health Pathogens**
Differential Equations Seminar, University of Maryland Baltimore County, Baltimore, Maryland

2025 **Modeling long-term viral infections**
Mathematics Seminar, Shippensburg University, Shippensburg, Pennsylvania

2025 **Scenario and forecast modeling for respiratory viruses**
Navajo Nation Epidemiology Center Data Summit, Window Rock, Arizona

2024 **Impacts of host behavior and viral life history on infectious disease transmission**
Department of Mathematics, Virginia Tech, Blacksburg, Virginia
Infectious Disease Dynamics Group, Johns Hopkins University, Baltimore, Maryland

2024 **Within-host viral life histories predict FMDV transmission dynamics for African buffalo**
Hawlena Lab, Ben-Gurion University, Midreshet Ben Gurion, Israel

2023–2024 **Modeling process-based effects of harmful algal blooms**
Department of Mathematics, University of Essex, Colchester, United Kingdom (2024)
School of Zoology Seminar, Tel Aviv University, Tel Aviv, Israel (2023)
Israel Oceanographic and Limnological Research, Haifa, Israel (2023)
Inter-University Institute for Marine Sciences, Eilat, Israel (2023)

2023 **Patterns of cultural variation in the Austronesian expansion**
Fortunato Lab, University of Oxford (virtual)

2022 **Consilience in FMDV ecology**
Gresham Lab, New York University, New York City, New York
Obolski Lab, Tel Aviv University, Tel Aviv, Israel

Selected Conference Presentations

2026 **Decision-Support Modeling for One Health Pathogens: Using Mechanistic Models for Surveillance and Forecast Design**
Society for Mathematical Biology Annual Meeting, Graz, Austria; mini-symposium talk

2025 **Recovering Ecological Geometry: A Trait- and Depth-Structured IPDE Model of Plankton Dynamics**
Society for Mathematical Biology Annual Meeting, Edmonton, Alberta, Canada; mini-symposium talk

2025 **Biologically Grounded Surrogate-Driven Parameter Inference for Sparsely Observed Dynamical Systems**
Symposium on Biomathematics and Ecology Education and Research (BEER), Fairfax, Virginia; contributed talk

2023 **Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms**
Society for Mathematical Biology Annual Meeting, Columbus, Ohio; mini-symposium talk

2022–2023 **Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo**
SIAM TXLA, Lafayette, Louisiana (2023); mini-symposium talk
SMB Epidemiology-Immunology Midyear Meeting (2022; virtual); contributed talk

2022–2023 **Dengue immunodynamics: antibody-dependent enhancement, cross-reactive immunity, and model–data confrontation**
SIAM TXLA, Lafayette, Louisiana (2023); mini-symposium talk
Mathematical Perspectives on Immuno-Biology, Blagoevgrad, Bulgaria (2023); contributed talk
Center for Combatting Pandemics Conference 2, Tel Aviv, Israel (2023); contributed talk
European Conference on Theoretical and Mathematical Biology, Heidelberg, Germany (2022); mini-symposium talk

2021 **Disease dynamics necessarily operate across scales**
SIAM TXLA, South Padre Island, Texas; mini-symposium talk

2021 **Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates**
Society for Mathematical Biology Annual Meeting (virtual); contributed talk

Teaching

Summer 2025 **Teaching Assistant**, *Johns Hopkins University*, Baltimore, Maryland
Teaching assistant for *Topics in Infectious Disease Epidemiology* with Dr. Shaun Truelove and Dr. Sara Loo; supported course delivery and applied instruction in infectious disease modeling and epidemiology.

2017–2020 **Teaching Assistant and Instructor of Record**, *University of Louisiana at Lafayette*, Lafayette, Louisiana
Instructor of record for elementary statistics (3 sections), survey of calculus (2 sections), precalculus (2 sections), and college algebra (2 sections). Teaching assistant for calculus, linear algebra, differential equations, and basic statistics.

2016–2017 **Teaching and Tutoring**
Instructor of record for North Carolina Math III / Algebra II, McDowell County Schools, Marion, North Carolina (2017). Mathematics and statistics tutoring at Memorial University of Newfoundland (2016–2017) and University of North Carolina Asheville (2016).

Mentoring, Outreach, and Service

Mentoring

2021–Present **Mentoring students and postdoctoral researchers in mathematics, statistics, and infectious disease modeling**
Jolles and Beechler Labs, Oregon State University (2025–Present)
Society for Mathematical Biology (2022–Present)
Association for Women in Mathematics (2023–2025)
Ram Lab, Tel Aviv University (2022–2024)
MIDAS Network (2021–2022)
Gulbudak Group, University of Louisiana at Lafayette (2021–2022)

Undergraduate Research Supervision

2024 **Shany Naim**, *Tel Aviv University*
Supervised undergraduate research project: initial translation of the VBPCApy variational Bayesian PCA codebase from MATLAB to Python.

Community Outreach

2021–2022 **Public engagement in science communication and research dissemination**
University of Louisiana R1 Showcase: selected to present COVID-19 modeling research at the university-wide showcase (2022)
Ocean View United Methodist Church, Oak Island, NC: community discussion on vaccines and COVID-19, with on-site pharmacist for vaccination (2021)

Professional Service

2023–2025 **Member**, *Society for Mathematical Biology Diversity, Equity, and Inclusion Committee*

2021–2023 **Session Chair / Leader**
SMB Early Career Workshop: Applying for postdoctoral funding (2023)
SMB Epidemiology-Immunology Midyear Meeting: Contributed talks (2022)
SMB Annual Meeting: CT04-MEPI contributed talks in mathematical epidemiology (2021)

2019–2023 **Mini-symposium Co-organizer**
SIAM TXLA: Data-driven mathematical and statistical modeling approaches to population biology (2023)
SMB Annual Meeting: Disease Dynamics Across Scales (2023)
SMB/ESTMB Joint Meeting: Multi-scale and data-driven modeling approaches in ecology, immunology, and epidemiology (2022)
SIAM TXLA: Reproducibility, reliability, and robustness: confronting models from across mathematical biology with data (2021)
SIAM TXLA: Population Dynamics in Ecology and Epidemiology (2019)

2021–2022 **Secretary**, *University of Louisiana AMS Graduate Student Chapter*

Journal Reviewing

2022–Present *PLOS ONE*, *Journal of Biological Systems*, *Journal of Biological Dynamics*, *Frontiers in Public Health*, *Frontiers in Veterinary Science*, *Computational and Applied Mathematics*, *Mathematical Biosciences*, *Scientific Reports*, *PLOS Computational Biology*, and *BMC Infectious Diseases*.

Honors and Awards

2019–2022 **Travel Awards**, *SIAM, Society for Mathematical Biology, and University of Louisiana Graduate Research Association*

2021 **Rhodes Outstanding Graduate Assistant Award**, *Department of Mathematics, University of Louisiana at Lafayette*, Awarded to two graduate students annually (approximately 50 graduate students total)

2021 **Outstanding Contributed Talk Award (Mathematical Epidemiology)**, *Society for Mathematical Biology Annual Meeting*

2019 **Three Minute Thesis Competition, Finalist**, *Louisiana Sea Grant*

Computational Expertise

Modeling Dynamical systems (ODE/PDE/stochastic); compartmental and agent-based models; multivariate data analysis and dimensionality reduction; stability, bifurcation, and sensitivity analysis; parameter identifiability and inverse problems.

Inference Bayesian hierarchical models; variational and simulation-based inference; data assimilation; uncertainty quantification; model calibration and validation; proper scoring rules and forecast evaluation.

Computing Python, Julia, C++, Stan, R, MATLAB. Scientific computing; high-performance numerical solvers; operator splitting; data pipelines and exploratory analysis; reproducible workflows (Git, CI/CD, containerization).

Professional Affiliations

Society for Mathematical Biology; Society for Industrial and Applied Mathematics; International Society for Bayesian Analysis; United States Research Software Engineer Association; MIDAS Network; Association for Women in Mathematics; Institute of Industrial and Systems Engineers; Society for Health Systems (SHS/IISE); and Healthcare Information and Management Systems Society (HIMSS).